

Classification of Deep Learning

1. Artificial Neural Networks (ANN)

- **Fundamental Idea:**

ANNs are inspired by the human brain and consist of interconnected neurons that learn patterns and relationships from data. They are general-purpose networks used for mapping inputs to outputs.

- **Structure:**

- Input Layer: Receives raw data (features).
- Hidden Layers: Perform computations using weighted sums and activation functions to learn patterns.
- Output Layer: Produces predictions or classifications.

- **Strengths:**

- Can model non-linear relationships.
- Flexible and used as a foundation for other deep learning architectures.

- **Applications:**

- Handwritten digit recognition
 - Basic image or text classification
 - Regression problems
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2. Convolutional Neural Networks (CNN) – mainly for Computer Vision

- **Fundamental Idea:**

CNNs automatically learn spatial hierarchies of features in images (edges → textures → shapes → objects) using convolution operations. This reduces the need for manual feature engineering.

- **Structure:**

1. Convolutional Layers: Apply filters/kernels to extract local patterns.
2. Pooling Layers: Reduce feature map size while keeping important features (e.g., max pooling).
3. Fully Connected Layers: Integrate all features for final decision-making.

- **Strengths:**

- Excellent for image and video data.
- Handles high-dimensional input efficiently.

- Learns features automatically.
 - Applications:
 - Image classification (e.g., detecting cats, dogs)
 - Object detection and segmentation
 - Facial recognition and emotion detection
 - Medical imaging (tumor detection, X-ray analysis)
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3. Recurrent Neural Networks (RNN) – mainly for Sequential Data

- Fundamental Idea:

RNNs are designed to handle sequences of data, maintaining a memory of previous inputs to capture temporal or contextual information.
- Structure:
 - Input at each time step → Hidden state (recurrent) → Output
 - Hidden state carries information forward, allowing the network to remember past inputs.
- Variants for better performance:
 - LSTM (Long Short-Term Memory): Handles long-term dependencies.
 - GRU (Gated Recurrent Unit): Simplified version of LSTM, faster to train.
- Strengths:
 - Good at modeling sequences, order, and temporal patterns.
 - Can process variable-length input sequences.
- Applications:
 - Natural language processing (text generation, translation, sentiment analysis)
 - Speech recognition and synthesis
 - Time-series forecasting (stock prices, weather prediction)
 - Handwriting recognition

Deep Learning Application Fields

- **Computer Vision (CV)**
- **Natural Language Processing (NLP)**

Computer Vision (CV) is a field of Artificial Intelligence that enables computers to see, interpret, and understand visual information from the world, such as images and videos. It allows machines to identify objects, faces, actions, and scenes using deep learning models like Convolutional Neural Networks (CNNs). Applications of computer vision include facial recognition, object detection, medical image analysis, and autonomous driving, where systems learn to extract meaning from visual data much like humans do.

Natural Language Processing (NLP) is a branch of Artificial Intelligence that helps computers understand, interpret, and generate human language in both text and speech. It combines linguistics and machine learning to enable machines to read, listen, and respond naturally. Using models like RNNs, LSTMs, and Transformers, NLP powers systems such as chatbots, voice assistants, machine translation, and sentiment analysis, allowing computers to communicate with humans in a meaningful way.